

THEORETICAL AND EXPERIMENTAL APPROACHES TO STUDY TSPO1 INVOLVEMENT IN ERYTHROPOIESIS AND CHOLESTEROL TRANSLOCATION

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ABSTRACT

The translocator protein (TSPO) is an outer mitochondrial membrane protein. In mammals, in addition to cholesterol translocation TSPO involved in various important functions such as apoptosis, steroidogenesis, porphyrin transport, and heme biosynthesis. TSPO1 is supposed to form complexes with many proteins such as voltage dependent anion channels (VDAC) to import cholesterol within mitochondria. However, its function in erythroid cells is still unclear. In TSPO, the most common polymorphism is rs6971 which is the substitution of Alanine to Threonine at position 147 (A147T). Moreover, A147T polymorphism reported to impact on radioligand binding affinity and biological functions, especially neurosteroid biosynthesis. Therefore, the present study is structured around two parallel objectives: to computationally study the impact of the A147T polymorphism on cholesterol translocation and to experimentally identify the protein partners and oligomeric states of human TSPO1 (hTSPO1) in erythroid cells. To accomplish the first objective, all-atom molecular dynamics (MD) simulations of hTSPO1 were performed in DOPC membrane environment, complemented with 14% cholesterol concentration in long timescales. Two simulation systems were prepared to characterize the variants impact from polymorphism: the wild-type (A147) and the mutant (T147) forms of hTSPO1, allowing comparative analysis resulting from this mutation. Our findings suggest the transmembrane architecture remained preserved in both variants, indicating the maintenance of global fold of protein. However, differences on cavity volume and small conformational changes were observed. Additionally, A147 variant showed more frequent cholesterol contacts and mutation also shifted the cholesterol binding preferences from TM2 to TM3. The cholesterol flip-flop events were more dominant in A147 variant mainly occurred via some polar contacts with cholesterol binding motifs at TM5. Towards the second objective, hTSPO1 was expressed in K562 cell lines and purified by nickel affinity chromatography. Our initial results showed the distinct oligomeric state of hTSPO1 and its interaction with UROD and ALAS1 which are the key enzymes in heme biosynthesis pathway. Further validation of these partners is currently in progress using mass spectrometry. To our knowledge, this is the first long-time scale MD simulation study exploring the A147T polymorphism in hTSPO. Furthermore, experimentally identified partners will be subjected to MD simulation providing detailed insights regarding cholesterol translocation in erythroid biology.

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